

ortho-Allylation of 1-Arylpyrazoles with Allyl Phenyl Ether via Iron-Catalyzed C–H Bond Activation under Mild Conditions

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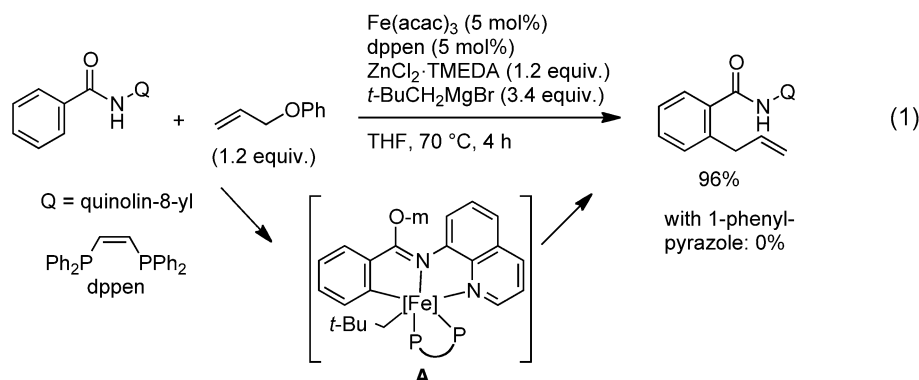
Abstract: An iron salt and a bipyridine-type ligand catalyze the *ortho*-allylation of 1-arylpyrazoles and congeners with allyl phenyl ether under mild conditions (0 °C). The ligand, an organozinc base, and the nature of the allylating reagent are crucial for the success of this reaction. Under these conditions, a competitive phenylation reaction is largely retarded, and cross-coupling of the organozinc with the allyl electrophile is minimized. The reaction may proceed *via* iron-catalyzed *ortho* C–H activation to form a metallic intermediate, which then reacts with the allyl ether in a γ selective fashion.

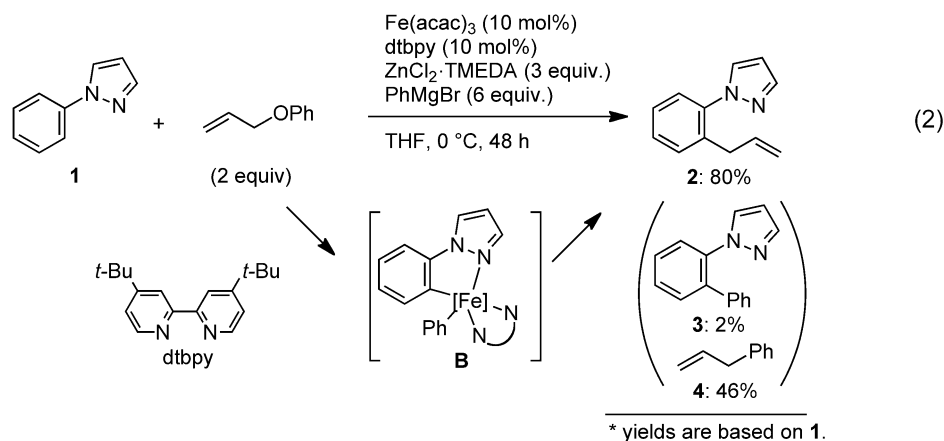
Keywords: allylation; arenes; C–H activation; iron; pyrazoles

the cleavage of the C–H bond and to control site selectivity.^[3] There has been much interest in diversifying the repertoire of directing groups, especially with the purpose of adding intrinsic value to the reaction products. The pyrazole group is a privileged structure of drug molecules and biologically active compounds;^[4] therefore, it has been an attractive choice as a directing group for C–H functionalization, typically *via* the use of precious metal catalysts or harsh reaction conditions.^[5] Herein, we report that an iron catalyst^[6] mediates the *ortho*-allylation of 1-arylpyrazole derivatives with allyl phenyl ether^[7] under remarkably mild conditions.^[8] Allylation of arenes through C–H activation^[9,10,11,12] is an attractive alternative to the Friedel–Crafts-type reaction,^[13] which is often limited to electron-rich arenes and suffers from selectivity issues.

The difficulties associated with the utilization of pyrazole as a directing group became apparent when we attempted to allylate 1-phenylpyrazole using the conditions developed previously in our laboratory for the allylation of aromatic carboxamides,^[14] which resulted in complete recovery of the starting material [Eq. (1)].

The formation of a carbon-carbon bond *via* transition metal-catalyzed C–H bond activation^[1] allows the straightforward creation of molecular complexity from simple hydrocarbons.^[2] A directing group that coordinates with the metal active species is usually appended to the hydrocarbon substrate, to facilitate





We ascribed this lack of reactivity to the poor compatibility of the pyrazole substrate with the ligand and organometallic base,^[15] resulting in the difficult formation of the putative metallacycle intermediate, such as **A**, that was proposed in our previous studies.^[14,16e,g] Therefore, we commenced an extensive study of these parameters (see the Supporting Information), which led to the finding that the use of a bipyridine-type ligand and diphenylzinc base is the key to the success of the allylation reaction, possibly by favoring the formation of an intermediate, such as **B** in Eq. (2). Thus, **1** reacted with allyl phenyl ether in the presence of 10 mol% of Fe(acac)₃/4,4'-di-*tert*-butyl-2,2'-bipyridyl (dtbpy) and 3 equiv. of a diphenylzinc reagent (prepared from ZnCl₂·TMEDA and PhMgBr) to give the *ortho*-allylated product **2** with a yield of 80% after isolation. We noted that the present catalytic system allowed the C–H activation reaction to proceed at 0 °C compared with the reaction indicated in Eq. (1), which required heating. Notably, under these conditions, only a trace amount of compound **3**, which was obtained *via* iron-catalyzed oxidative arylation^[16] with allyl ether as the oxidant, and 46% of **4** (based on **1**, 23% based on the initial allyl ether), which was obtained *via* iron-catalyzed allylic cross-coupling,^[17] were generated. As described in the Supporting Information, other ligands and bases resulted mainly in the recovery of the starting material. Sterically hindered MesMgBr or electron-deficient C₆F₅MgBr completely stopped the reaction.

An important factor for the success of this reaction was the choice of the allylating reagent (Figure 1). Thus, allylic electrophiles bearing a poor leaving group, such as thiolate, amide, and siloxy were mostly recovered, whereas those bearing a better leaving group, such as acetate, benzoate, carbonate, or chloride, reacted mostly with diphenylzinc to give **4**.

Various 1-arylpyrazoles and congeners could be allylated with allyl phenyl ether under the optimized conditions (Table 1). All the reactions gave the mono-allylated product selectively, and the diallylated prod-

2 : 65% (80%)	2 : 20%	2 : 2%	2 : 14%
4 : 17% (46%)	4 : 3%	4 : 1%	4 : 2%
2 : 19%	2 : 6%	2 : 1%	2 : 0%
4 : 155%	4 : 104%	4 : 177%	4 : 138%

Figure 1. Investigation of several allylating reagents (yields are based on **1**). The reaction time was 15 h, except for the yields in parentheses, for which the reaction time was 48 h.

uct was observed in a trace amount (<2%). This selectivity was in agreement with the poor reactivity of an *ortho*-substituted substrate (entry 7), which we ascribed to the steric interactions that disturb the formation of a planar metallacycle.^[16g] Isomerization of the allylated product to the more thermodynamically stable styrene derivative, which is sometimes an issue in the allylation reaction, was largely retarded (<5%). We observed though that a prolonged reaction time led to an increase in the amount of the styrene derivative.^[18] The phenylated product (such as **3**) was obtained in a trace amount, with the exception of electron-deficient substrates and 2-naphthylpyrazole (entries 5, 8, and 10), where it was obtained with a yield of 5–7%. The reaction did not reach complete conversion and the starting material was recovered, probably due to catalyst deactivation by the phenoxide anion that is generated as the reaction progresses.

The reaction of substrates bearing an electron-donating group at the *para*-position proceeded smoothly (entries 2–4), whereas a *para*-substituted substrate bearing an electron-withdrawing group was less reactive (entry 5). The allylation of *meta*-substituted substrates took place selectively at the less hindered position with good yields (entries 6 and 8). A substrate in which the pyrazole bears a methyl group reacted more efficiently than did a substrate in which the pyrazole bears an ester group (entries 9 and 10). We note that the ester group was untouched under the reaction

Table 1. Iron-catalyzed reaction of various 1-arylpyrazoles and congeners with allyl phenyl ether.^[a]

Entry	Substrate	Product	Time [h]	Yield [%] ^[b]
1 ^[c]			48	80 (X = H)
2			36	67 (X = Me)
3			18	60 (X = OMe)
4			48	64 (X = NMe ₂)
5			15	23 (X = F) ^[d]
6 ^[e]			48	65
7			48	9 ^[f]
8			24	54
9			24	71 (X = Me)
10			48	32 (X = CO ₂ Et) ^[d]
11			36	72

^[a] The reaction was performed under the conditions described in Eq. (2) on a 0.4-mmol scale. See the Supporting Information for details.

^[b] Yield for the isolated product.

^[c] On a 0.8-mmol scale.

^[d] Containing a trace amount of phenylated product.

^[e] ZnBr₂·TMEDA was used instead of ZnCl₂·TMEDA.

^[f] Determined by ¹H NMR.

conditions and that the unreacted substrate was recovered (entry 10). An indazolyl group, found in the structure of alkaloids such as nigellicine, nigelaniline, and nigellidine, also acted as an excellent directing group (entry 11). A further example of the peculiar nature of pyrazole as a directing group is the reaction

of an arene possessing a commonly used directing group such as pyridine, imine, *N*-methanamide, or triazole, which gave mainly the phenylated product (congener of **3**) and recovery of the starting material (see the Supporting Information).

Substituted allyl ethers reacted sluggishly under these reaction conditions and the arylpyrazole substrate was partially phenylated.

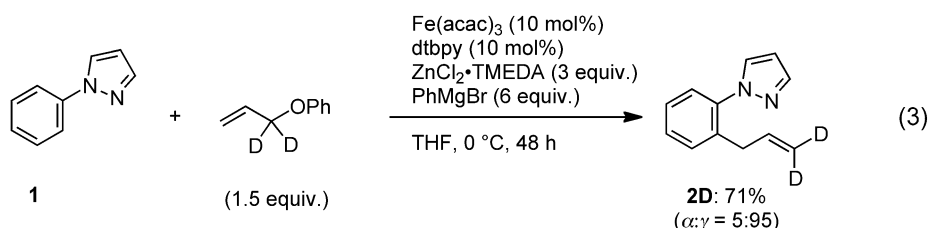
To gain a mechanistic insight into the allylation step, **1** was reacted with 1,1-dideuterioallyl phenyl ether [Eq. (3)]. The two deuterium atoms were found selectively at the terminal vinylic positions, suggesting that the allylation took place with high γ selectivity. Benzyl phenyl ether and homoallyl phenyl ether did not react at all, suggesting that the reaction proceeds through the insertion of the alkene with the necessary presence of a β leaving group. Involvement of an iron-catalyzed C–H activation step is supported by the following experimental facts: (i) the reaction in the absence of an iron catalyst gave no allylated product; (ii) the reaction in the presence of one equiv. of iron and without the allyl ether, followed by quenching with deuterium oxide resulted in deuterium incorporation at the *ortho*-position (see the Supporting Information); (iii) the same reaction performed in the absence of iron resulted in 0% deuterium incorporation.

In conclusion, we have developed an iron-catalyzed allylation of 1-arylpyrazole and congeners with allyl ethers under mild reaction conditions. This C–H activation reaction exploits the medicinally relevant pyrazole moiety as a directing group to create a new class of allylated pyrazole derivatives. Iron catalysis has been a topic of great interest because of the economic and environmental merits of iron,^[19] the present work represents a rare example of iron-catalyzed C–H activation followed by reaction with an electrophile.^[14,20] Further investigation of this reactivity of iron, as well as the expansion of the scope of the allylic substrate, will be the subject of our future studies.

Experimental Section

Typical Procedure

1-Phenyl-1*H*-pyrazole (**1**, 116 mg, 0.8 mmol) and ZnCl₂·TMEDA (606 mg, 2.4 mmol) were placed in an oven-



dried Schlenk flask under argon. A solution of PhMgBr in THF (1.09 M, 4.40 mL, 4.8 mmol) was added dropwise to this mixture at 0 °C. After stirring for 3 min, allyl phenyl ether (220 µL, 1.6 mmol) and a solution of Fe(acac)₃/4,4'-di-*tert*-butyl-2,2'-bipyridyl in THF (1.0 mL, 0.08 M, 80 µmol) were added sequentially. The reaction mixture was stirred at 0 °C for 48 h, followed by dilution with Et₂O and addition of a saturated aqueous solution of Rochelle's salt. After extraction with ethyl acetate, the combined organic layers were passed through a pad of Florisil and concentrated under reduced pressure. The crude product was purified by column chromatography on silica gel (hexane/AcOEt/NEt₃ = 99/0.5/0.5), followed by gel-permeation chromatography (eluent = toluene) to remove **1**, to afford 1-(2-allylphenyl)-1*H*-pyrazole (**2**) as a colorless oil; yield: 118 mg (80%). The spectroscopic data were in agreement with those reported in the literature.^[9d]

Acknowledgements

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